

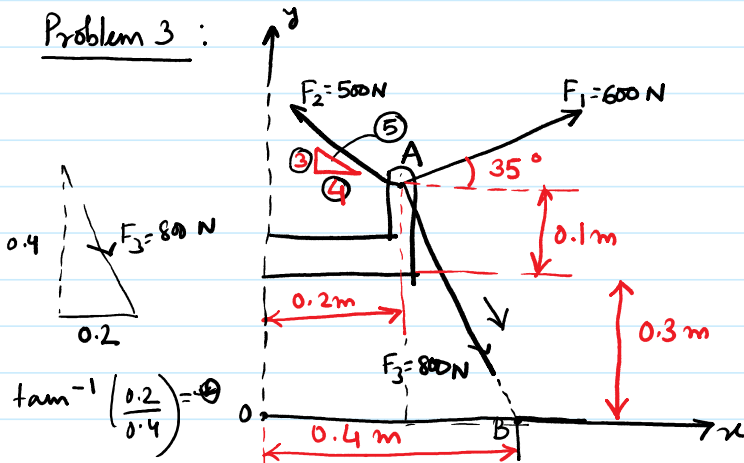
The 1800 N force F is applied to the end of I beam. Express F as a vector using unit vector $\hat{i}$ and $\hat{j}$.

$$F_x =$$

$$F_y =$$

$$F = -1080\hat{i} - 1440\hat{j} \text{ N}$$

Problem 3 :



Forces F_1 , F_2 and F_3 act on point A of the bracket. Determine x and y scalar components of each of the 3 forces.

$$F_{1x} = 600 \cos 35$$

$$F_{1y} = 600 \sin 35$$

$$F_{2x} = -500 \left(\frac{4}{5}\right) = -400 \text{ N}$$

$$F_{2y} = 500 \left(\frac{3}{5}\right) = 300 \text{ N}$$

$$F_{3x} = F_3 \sin \theta$$

$$F_{3y} = -F_3 \cos \theta$$

n_{AB}

B (0.4, 0)

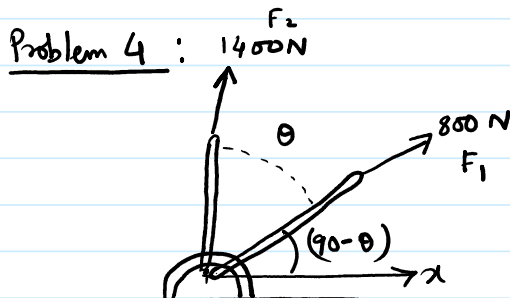
$\vec{AB}$

A (0.2, 0.1)

n_{AB}

800 n_{AB}

F_{3x}
 F_{3y}



At what angle θ must 800 N force be applied in order that the resultant R of the two forces have a magnitude 2000 N? For this condition, determine angle β between R and vertical.
 $R = 2000 \text{ N}$.

$$F_{1x} = 800 \sin \theta$$

$$F_{2x} = 0$$

$$F_{1y} = 800 \cos \theta$$

$$F_{2y} = 1400$$

$$\text{Ans: } \theta = 51.3^\circ$$

$$\beta = 18.19^\circ$$

$$\left. \begin{aligned} R_x &= F_{1x} + F_{2x} = 800 \sin \theta \\ R_y &= 800 \cos \theta + 1400 \end{aligned} \right\} \begin{aligned} R &= \sqrt{R_x^2 + R_y^2} \\ 2000 &= \sqrt{R_x^2 + R_y^2} \end{aligned}$$

$$R_y = 800 \cos \theta + 1400$$

)

$$2000 = \sqrt{R_x^2 + R_y^2}$$

$$\tan^{-1} \left(\frac{R_y}{R_x} \right) = \text{angle of } R \text{ w.r.t } x\text{-axis} = \alpha$$

$$\beta = 90 - \alpha$$

Problem 5 :